

- 35) A bounce rate is the percentage of visitors who leave a website after viewing only one page, without interacting (click, filling a form) or navigating further. It ~~is~~ indicates how well a page engages visitors or meets their needs, with a high percentage often highlighting poor content relevance, slow page load speeds, or bad mobile usability.
- 36) A/B testing, or split testing, is a controlled experiment that compares two versions of a digital asset - a control (A) and a variation (B) to determine which performs better. Traffic is split equally to measure, analyze, and select the winning version based on key metrics like conversions, clicks, or engagement.
- 37) Lead generation is the strategic process of identifying, attracting & capturing potential customers contact information to nurture them into paying clients. It bridges marketing & sales by creating interest through content, social media & ads, utilizing tools like landing pages, forms & gated content to initiate engagement.
- 38) Marketing Attribution determines which channels such as social-media, search, or email drive conversions, allowing for optimized budget allocation. No single channel always deserves credit, rather, credit depends on the model used (e.g. last click, first-touch, linear), with multi-touch models often providing the most accurate picture of complex customer journeys.

- 31) Incremental Revenue is the additional income a business generates as a direct result of a specific action, such as a marketing campaign promotion, or new product launch, above and beyond the "baseline" revenue they would have earned anyway. It measures the "lift" or true causal impact of a campaign rather than just total revenue, which includes sales that would have occurred without any marketing effort.
- 32) Return on Investment (ROI) is a fundamental financial metric used to evaluate the efficiency, profitability or potential gain of an investment relative to its cost. It is widely considered the "Classic" profitability test because it provides a quick, standardized percentage that allows for easy comparison between different types of investments or projects
- 33) Return on Ad Spend (ROAS) is a marketing metric measuring gross revenue generated for every dollar spent on advertising. Calculated as $\text{revenue} \div \text{cost}$. It determines campaign profitability & effectiveness, often expressed as a ratio (e.g. 4:1 or \$4) or percentage (e.g. 400%). A typical baseline is 4:1, though it varies by margin
- 34) Click-Through Rate (CTR) measures the percentage of people who click a link, ad, or email after seeing it, calculated as $(\text{Clicks} \div \text{Impressions}) \times 100$. It determines audience engagement, with a typical "good average" falling between 2% & 5%, though this varies widely by industry, platform & content

- 26) Net Promoter Score (NPS) measures customer loyalty and likelihood to recommend a brand on a scale of 0-10. Calculated by subtracting the percentage of detractors (0-6) from promoters (9-10), it helps companies gauge customer satisfaction, identify brand advocates, & forecast growth, with scores over 50 generally considered ^{excellent}.
- 27) Average Order Value (AOV) measures the average revenue generated per transaction, calculated by dividing total revenue by the number of orders. It is vital ecommerce metric for tracking customer purchasing behavior, measuring marketing effectiveness and improving profitability without increasing acquisition costs.
- 28) A sales funnel is a visual representation of the customer journey, mapping the path from the moment a consumer becomes aware of a brand to the final purchase. It is structured as a funnel because a large number of potential customers (leads) enter at the top, but only a smaller more qualified percentage move through to the bottom to become paying customers.
- 29) A marketing campaign is a structured, time bound and coordinated effort designed to achieve specific business goals such as launching a product, increasing brand awareness, or driving sales. It unites multiple channels (eg email, social media, ads) to deliver a consistent message, often proving 31% more effective than standalone efforts.
- 30) Post-campaign analysis evaluates marketing performance to determine ROI, audience resonance, and strategic effectiveness. It involves comparing KPIs such as ROAS, conversion rates & engagement against initial goals analyzing channel performance and documenting lessons learned. This process turns data into actionable insights for future optimization.

22) Customer Acquisition Cost (CAC) is the total expense including marketing, sales and overhead required to convince a potential customer to buy a product or service. It is a critical metric for determining profitability & growth sustainability, calculated as $\frac{\text{Total Acquisition Costs}}{\text{Total New Customers}}$ over a specific period.

23) Customer Lifetime Value (CLV or LTV) is a metric that predicts the total net profit or revenue a business can expect from a single customer throughout their entire relationship. It helps identify high value customers, guide marketing spend, and optimize retention strategies, with a healthy CLV often being 3x the Customer Acquisition Cost (CAC).

24) Churn rate is the percentage of customers or subscribers who cancel or stop using a company's product/service over specific period. It is a key metric for business health, typically calculated as: $(\text{Customer lost} / \text{Total Customer at Start}) \times 100$. High churn indicates poor retention or dissatisfaction, while low churn indicates high satisfaction.

25) Customer retention rate measures the percentage of existing customers who remain with a company over a specific period, serving as the direct inverse of the churn rate. It indicates customer loyalty, satisfaction, and the effectiveness of retention strategies, with high retention generally correlating to low, healthy churn rates.

18) Key Performance Indicators (KPIs) are quantifiable, high-level metrics used to measure an organization's success in achieving strategic, financial & operational goals. They act as a "GPS", helping businesses evaluate performances, make data driven decisions, & ensure efforts align with objectives. Effective KPIs are typically "Smart" (Specific, Measurable, Attainable, Relevant, Time-bound).

19) A benchmark is a standard, point of reference, or criterion against which products, services processes or performance can be measured, compared & evaluated. It acts as a foundational tool for improvement, allowing organizations to compare their current standing against competitors, industry best practices, or their own previous performance to identify gaps & areas for growth.

20) Customer segmentation is the process of dividing a customer base into distinct, actionable groups based on shared attributes like demographics, behaviours & needs. It enables personalized marketing, improved customer experience & higher retention by allowing businesses to tailor products & messaging to specific rather than generic groups.

21) A Conversion rate is the percentage of website visitors who complete a desired action such as making a purchase, signing up for a news letter, or filling out a form. It is a vital metric for measuring marketing effectiveness & website performance, typically calculated by dividing the number of conversions by the total number of ^{visitors} ~~visitors~~ then multiplying by 100.

13) A data warehouse is a central repository that aggregates structured data from multiple, disparate sources (such as CRM, ERP, and SQL databases) to support analytics, reporting, and business intelligence (BI). It is designed specifically for complex querying & trend analysis rather than running daily transactions.

14) Data Mining is the process of analyzing large datasets to uncover hidden patterns, trends and correlations, turning raw data into actionable knowledge. Using statistics and machine learning, it extracts meaningful information for decision-making across industries like finance & healthcare. Common patterns include associations, clusters and predictive models.

15) Data Visualization is the graphical representation of information & data, using visual elements like charts, graphs, maps & infographics to make complex data sets accessible, understandable, & actionable. It highlights trends, patterns, and outliers, enabling faster insights, better decision-making & improved story telling.

16) Dashboard - A live performance dashboard provides a real-time, visual and centralized view of key business metrics. It allows teams to track performance and make immediate, data-backed decisions without waiting for end-of-week reports.

17) Metric - The metric system is a standardized, decimal-based system of measurement used worldwide, officially known as the International System of Units (SI). It is designed to measure, quantify & standardize physical properties like length, weight and volume, expressing them as numbers based on powers of 10.

4) Prescriptive analytics in the final, most actionable stage of business analytics that answers the question "What should we do about it?" by recommending specific, optimized actions to achieve desired outcomes. It goes beyond forecasting (predictive) to suggest best-case scenarios & actions using AI, data, and algorithms, such as automating supply chain adjustments or personalized customer retention tactics.

10) Data literacy the ability to read, work with, analyze & communicate with data is a foundational 21st century skill crucial for navigating a data-driven world. It empowers individuals to convert raw data into actionable insights, driving informed decisions, increasing efficiency, & fostering competitive advantage across all industries.

11) Structured data is highly organized information that fits into fixed fields, rows and columns, adhering to a predefined schema. It is easily searchable & analyzable by machines & humans using SQL, often used for customer information, financial transactions, and web SEO to improve search engine understanding.

12) Unstructured data refers to information that does not have a predefined data model or organizational structure, making it difficult to store in traditional relational databases (row & columns). It is generally qualitative, messy, and accounts for over 80-90% of all data created today.

5) Big Data is characterized by three core dimensions volume, velocity and variety (the "3 Vs") - Which define the challenges & opportunities of handling massive data sets. These characteristics represent the immense amount of data, the high speed of its generation & processing, and the diverse formats that cannot be handled by traditional database systems

6) Descriptive analytics is the foundational process of summarizing historical and current data to understand past performance, directly answering the question "What happened?". It uses techniques like data aggregation, data mining, and visualization (reports, dashboards) to identify trends, patterns, and KPIs, such as monthly sales, website traffic, or churn rates.

7) Diagnostic analytics answer "Why did it happen?" by examining historical data to identify the root causes, patterns, and correlations behind specific past events. It bridges the gap between knowing what happened (descriptive) & predicting what might happen (predictive) by, for example, finding why sales dropped or why customer churn spiked

8) Predictive analytics uses historical data, machine learning, and statistical algorithms to forecast future outcomes, directly answering, "What is likely to happen?". By identifying patterns in past behavior & data, it allows businesses to anticipate trends, reduce risk, and act proactively rather than reacting to events after they occur.

The language of Business Analytics & Terminology & Research

- 1) Business Analytics (BA) is a disciplined, data-driven approach and analytics technology, statistical analysis, and quantitative methods to examine business performance, derive actionable insights, and guide decision-making. It is a subset of business intelligence (BI) that acts as a bridge between data science & strategic business decisions.
- 2) Data Analytics is the science of analyzing raw data using specialized tools and processes to uncover patterns, trends & insights. It transforms data into actionable information, enabling businesses to make informed, evidence-based decisions, optimize efficiency, and predict future outcomes. It involves collecting, cleaning and visualizing data to answer specific questions.
- 3) Business Intelligence (BI) tools transform raw data into actionable insights, providing reporting, data visualization, and analytics to support strategic decision-making. These tools connect to various data sources such as CRM, ERP, and spreadsheets to create interactive dashboards that help business reduce inefficiencies, identify new revenue streams and track key performance indicators (KPIs).
- 4) Data-driven decision-making (DDM) is a modern management philosophy that prioritizes evidence-based analysis over intuition, using collected data to guide strategic, operational & financial decisions. It enhances accuracy, reduces risks, and fosters a proactive, competitive organizational culture.